Q30: Project Suggestions

DLS 2026 — Kukla

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1. **Measure graph breaks (small, 2–3 days).** Take a model with data-dependent control flow; use `torch._dynamo.explain` to count graph breaks, refactor to remove some, and measure the speedup from `torch.compile`.

2. **Checkpointing trade curve (medium, 3–5 days).** On a deep net, sweep how many layers are gradient-checkpointed; plot peak activation memory vs. step time to map the compute–memory trade.

3. **Tape vs. functional memory (medium, 1 week).** Implement the same model with `autograd` and with `torch.func`; measure peak memory and discuss where (and why) they differ, and whether checkpointing closes the gap.